Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think> The de Broglie wavelength can be calculated using the formula Î» = h / (sqrt(2me * K)), where h is Planck's constant, me is the mass of the electron, and K is the kinetic energy. Converting 1eV to joules (1eV = 1.602 ÃĹ10^-19 J) and knowing the mass of an electron (approximately 9.109 ÃĹ10^-31 kg), and Planck's constant (6.626 ÃĹ10^-34 Js), I can plug in these values to find the wavelength. After simplifying, I get around 1.23 ÃĹ10^-7 cm.

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</think><answer> G. 1.23 ÃĹ10^-7 cm </answer>